

AP AUTOMATION

Intelligent Invoice Coding for Finance Teams

From manual GL coding to self-learning automation —
faster processing, stronger governance, measurable ROI.

September 2026

The manual AP problem



Slow

Clerks manually code every non-PO invoice. Each one takes 2–5 minutes. Backlogs build.



Error-prone

Manual coding produces inconsistent GL assignments. Corrections require rework and audit trails.



Expensive

AP clerk time is consumed by repetitive coding. Scale requires more headcount, not smarter systems.

How the system works

1

Invoice arrives

Validated, vendor resolved,
description cleaned

2

RAG retrieval

Similar historical invoices
retrieved from the corpus

3

Confidence gate

High confidence → auto-
post. Low confidence →
human review

4

Write-back

Every confirmation trains the
system. It gets better
automatically

Proven POC results

99%

Prediction accuracy

198/200 invoices correct

74%

Straight-through rate

Zero human touch needed

100%

Auto-post accuracy

Never wrong when
confident

907

Corpus points

Growing with every
confirmation

Results from live POC running on Qdrant Cloud with real embedding model — not simulated.

Time to value

Week 1–2

Historical invoice extract from ERP seeds the corpus. Shadow mode begins — system runs alongside existing process.

Week 3–4

Accuracy validated against confirmed postings. Confidence thresholds calibrated per vendor category.

Week 5–6

Go live. Auto-posting begins. Clerks handle exceptions only. Corpus grows with every confirmation.

Month 3+

Automation rate climbing. New vendor types brought in. Reporting suite live for Finance Director.

The savings case

AT 10,000 INVOICES/MONTH

AP clerk cost (manual)	~£2,300/month
System infrastructure	~£300/month
Automation rate	74% straight-through
Net monthly saving	~£1,900/month

BEYOND THE NUMBERS

- ✓ Clerks redeployed to exceptions and higher-value work
- ✓ Invoices posted same day — not end of week
- ✓ Scale without headcount — double volume, add £50/month
- ✓ ROI compounds as automation rate climbs over time

Governance by design

Confidence gate

System only auto-posts when it is confident. Low confidence always routes to human review. The gate is never switched off.

Cold-start protection

Unknown vendors with no history are never auto-posted. They go to review until enough evidence is built up.

High-value override

Invoices above a defined threshold always require human approval — regardless of confidence score.

Full audit trail

Every prediction carries full provenance — retrieved evidence, confidence score, routing reason, confirming user. Exportable for external audit.

Immutable log

Append-only, cryptographically hashed. No decision can be altered after it is recorded. Full reconstruction from a single trace ID.

The principle

Governance is proportional, not absent. The system scales oversight with risk — it does not remove it.

The system gets better without anyone touching it

Traditional AP automation requires a professional services team to author and maintain rules. Every new vendor, every coding policy change, every ERP update — manual intervention.

This system learns from every confirmed invoice. The corpus grows. Confidence scores improve. Automation rate climbs. No rules. No PS bottleneck. No ongoing cost.

Month 1 — seed corpus	60% auto-post
Month 3 — corpus growing	74% auto-post
Month 6 — well established	85%+ auto-post
Year 2 — mature corpus	90%+ target

Reporting your ERP cannot provide

AP Team

- Queue depth and age
- Automation rate trend
- Cold-start vendor list
- Exception report

Finance Director

- Real-time spend by GL / cost centre
- Variance analysis per vendor
- Currency exposure
- Accruals support

Compliance

- Full decision log per invoice
- Threshold change log
- Override report
- High-value invoice audit trail

CFO / COO

- ROI dashboard — hours saved
- Automation rate by entity
- Accuracy trend
- Corpus growth month on month

The ERP sees what was posted. This system sees why, how confident, and whether a human intervened.

Five things competitors do not have together

Self-learning

Write-back loop compounds automatically. No PS team needed.

Proportional governance

Confidence gate scales oversight with risk. Never removes it.

Auditability by design

Full provenance per decision. Not bolted on after the fact.

Agentic reasoning

LLM explains ambiguous decisions in plain English. Clerks understand why.

Multi-tenant isolation

Enterprise-safe from day one. Every layer isolated per client.

AP AUTOMATION

From manual coding to self-learning automation. In six weeks.

£1,900

monthly saving per 10k invoices

99%

prediction accuracy (proven)

6 weeks

to production v1

Live POC available to demo — ap-automation-tjfgztux7pwux2mu8v7ewj.streamlit.app